TAI NGUYEN PHU

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Final-year Computer Science student with a solid foundation in Machine Learning and NLP, currently specializing in Large Language Models (LLMs), semantic retrieval, and generative AI agents. Seeking an AI/NLP Engineer internship or entry-level role to contribute to innovative solutions and help build scalable, real-world applications powered by LLMs and retrieval-augmented generation.

EDUCATION

Bachelor of Computer Science — **Sep 2022 - Jun 2025**
University of Information Technology - VNUHCM, Vietnam — **GPA:** 8.24/10

CERTIFICATIONS

IELTS Academic (Overall: 7.0) — **Feb 2022**

PROJECTS

Vietnamese Legal Document Retrieval — **Apr 2025 - Apr 2025**

Technologies: Python, HuggingFace, SentenceTransformers, Gradio, Docker, Pandas, FAISS

- Fine-tuned a Sentence-BERT model (bert-base-multilingual-cased) on a 100K+ Vietnamese legal documents using CachedMultipleNegativesRankingLoss for domain-specific semantic retrieval.
- Achieved significant performance boost on MTEB evaluation benchmark (NDCG@10: 60.4%, MAP@10: 53.6%) compared to pre-trained baseline (0.023% / 0.014%).
- Designed and deployed a scalable semantic search pipeline using FAISS for real-time legal document retrieval.
- Built a user-friendly web interface with Gradio to facilitate interactive query testing.
- Containerized the full system with Docker to ensure reproducibility, portability, and seamless deployment.

Semantic Book Recommender — **Apr 2025 - Apr 2025**

Technologies: Python, HuggingFace Transformers, Gradio, Pandas, NumPy

- Developed an AI-powered book recommendation system that leverages semantic search and zero-shot classification to suggest books based on user input.
- Utilized pre-trained transformer models to generate embeddings for book descriptions, enabling semantic similarity comparisons.
- Implemented zero-shot classification to categorize books into genres without labeled training data.
- Built an interactive web application using Gradio, allowing users to input descriptions and receive personalized book recommendations.
- Conducted exploratory data analysis and sentiment analysis to enhance recommendation relevance.

Job Application AI Assistant — **Mar 2025 - Mar 2025**

Technologies: Python, Llama, LangChain, FAISS, Streamlit

- Developed a Gen AI assistant to streamline job applications by automating 80% of the cold email generation process and providing a tailored chatbot for job application guidance based on job postings and applicant resumes.
- Implemented FAISS-based semantic search for efficient retrieval of job descriptions and resumes.
- Integrated LangChain to enhance AI-driven responses with contextual understanding.
- Designed and deployed a Streamlit UI, ensuring seamless user interaction.

Scene Text Recognition — **Feb 2025 - Feb 2025**

Technologies: Python, PyTorch, YOLOv11m, CRNN, FastAPI, Ray Serve, Streamlit, OpenCV

- Developed a Scene Text Recognition (STR) pipeline using YOLOv11m for text detection and CRNN (ResNet34) for recognition, achieving ~88% precision on the ICDAR2003 dataset.
- Developed a scalable OCR API with FastAPI and Ray Serve, enabling autoscaling, GPU acceleration, and real-time text extraction.
- Created an interactive Streamlit web application, allowing image processing via file upload or URLs for instant text recognition.
- Implemented transformer-based text processing and CTC loss for sequence modeling, ensuring accurate recognition of text in natural scene images.

Data Science Job Salary Prediction — **Dec 2024 - Jan 2025**

Technologies: Python, Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn, Streamlit

- Developed a machine learning-powered salary prediction model for Data Science jobs, trained on real-world Glassdoor job market data.
- Engineered and optimized features using EDA, one-hot encoding, label encoding, and feature importance analysis.
- Trained and compared multiple ML models (XGBoost, Random Forest, Decision Tree, Gradient Boosting, Linear Regression, SVM), achieving $R^2 = 0.815$ with XGBoost.
- Implemented hyperparameter tuning with GridSearchCV & RandomizedSearchCV to maximize prediction accuracy.
- Built and deployed an interactive Streamlit web application, allowing users to input job details and receive salary predictions.

SKILLS

Languages: Python, SQL, C++

Frameworks: PyTorch, Transformers, SentenceTransformer, LangChain, FAISS, OpenCV, Numpy, Pandas, Matplotlib, SciKit-Learn

Deployment: Docker, FastAPI, Streamlit, Gradio, Ray Serve

Soft Skills: English, Presentation, Critical Thinking, Teamwork, Problem-solving